

The Effect of In-Store Travel Distance on Unplanned Spending: Applications to Mobile Promotion Strategies

Typically, shoppers' paths only cover less than half of the areas in a grocery store. Given that shoppers often use physical products in the store as external memory cues, encouraging shoppers to travel more of the store may increase unplanned spending. Estimating the direct effect of in-store travel distance on unplanned spending, however, is complicated by the difficulty of collecting in-store path data and the endogeneity of in-store travel distance. To address both issues, the authors collect a novel data set using in-store radio frequency identification tracking and develop an instrumental variable approach to account for endogeneity. Their analysis reveals that the elasticity of unplanned spending on travel distance is 57% higher than the uncorrected ordinary least squares estimate. Simulations based on the authors' estimates suggest that strategically promoting three product categories through mobile promotion could increase unplanned spending by 16.1%, compared with the estimated effect of a benchmark strategy based on relocating three destination categories (7.2%). Furthermore, the authors conduct a field experiment to assess the effectiveness of mobile promotions and find that a coupon that required shoppers to travel farther from their planned path resulted in a substantial increase in unplanned spending (\$21.29) over a coupon for an unplanned category near their planned path (\$13.83). The results suggest that targeted mobile promotions aimed at increasing in-store path length can increase unplanned spending.

Keywords: shopper marketing, path data, radio frequency identification tracking, unplanned purchase, mobile promotion

Recent studies of in-store shopping behavior (Hui, Bradlow, and Fader 2009; Hui, Fader, and Bradlow 2009b; Hui et al. 2012; Larson, Bradlow, and Fader 2005) have suggested that consumers rarely shop the entire store. Contrary to the conventional wisdom that shoppers go through a grocery store aisle by aisle, Larson, Bradlow, and Fader (2005) and Hui, Fader, and Bradlow (2009) find that shoppers typically walk through the perimeter of the store and only visit the specific aisles they need. As a result, on average, shoppers only visit approximately one-third of all store areas (Hui and Bradlow 2012). Thus, for each

shopper, the majority of store areas are bypassed, and the product categories in those areas remain unseen.

Consistent with the industry adage that "unseen is unsold," research consistently reports that shoppers often use physical products in the store as external memory cues that create new needs or triggers forgotten needs (Inman and Winer 1998; Kollat and Willett 1967; Park, Iyer, and Smith 1989). Thus, strategies that encourage shoppers to travel more of the store may increase unplanned spending by exposing shoppers to more product stimuli during their shopping trips. Examples include the "classic" strategy of scattering popular product categories (e.g., milk, eggs) around the store (Granbois 1968; Iyer 1989) and the emerging technology of using mobile promotions to entice shoppers to visit more store areas (e.g., www.motomessage.com). The effectiveness of these in-store shopper marketing strategies aimed at increasing in-store travel distances hinges on a causal, substantive effect of longer in-store trip distance on unplanned purchasing. If such a direct effect is both statistically and economically significant, strategies that increase travel distances will potentially lead to greater revenues from unplanned purchases.

Surprisingly, to our knowledge, there has been no research that explicitly studies the causal relationship between in-store trip length and unplanned spending. This gap in the literature is mainly due to (1) lack of data avail-

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ability and (2) the methodological challenges caused by the endogeneity of in-store path. First, before the recent development of radio frequency identification (RFID) tracking, in-store trip length was extremely difficult and costly to obtain (Hui, Fader, and Bradlow 2009a). Researchers had to either physically track shoppers (Farley and Ring 1966; Granbois 1968; Heller 1988) or rely on shoppers' self-reports (Inman, Winer, and Ferraro 2009), both of which may be inaccurate. Second, even if in-store trip length can be measured accurately, the causal effect of in-store path length still cannot be determined directly due to endogeneity issues resulting from omitted in-store and out-of-store variables, simultaneity/reversed causality, and measurement errors.

To fill this gap in the literature, we collected a novel data set using RFID tracking in conjunction with an entrance and exit survey. Specifically, we tracked each shopper using an RFID tag that enabled us to accurately measure their in-store path length. To account for endogeneity of the store path, we devised a novel instrumental variable approach (Greene 2007). We constructed an instrument based on the length of a "reference path," which is determined by the store layout, a shopper's planned purchases, and an assumption about his or her search strategy (infinitely forward looking [traveling salesman problem, or TSP] or one-step-look-ahead [ISLA]). We show that the lengths of the reference paths are strongly correlated with the lengths of the actual in-store paths. Importantly, because the reference paths are determined *before* the shopper starts a grocery trip, these instruments temporally precede the dependent variable (unplanned spending), thus allowing us to resolve the endogeneity issues.

Using the instrumental variable approach, we are able to estimate the direct effect of in-store path length on unplanned spending using two-stage least squares (2SLS; Greene 2007). We find that the elasticity of unplanned spending on in-store travel distance is approximately 1.57, which is 57% higher than the corresponding ordinary least squares (OLS) estimate that does not account for endogeneity. To put this into managerial perspective, for the shoppers in our study, increasing path length by 10% for each shopper (an average of approximately 140 feet) would increase unplanned spending by about 16.1%, or \$2.54 per shopper.

Our results and modeling framework enable us to assess the effectiveness of mobile promotion strategies in increasing unplanned purchases against the classic strategy of manipulating the store layout. We find that while relocating three product categories may increase unplanned spending by approximately 7%, strategically promoting three additional product categories using mobile promotion can increase the overall amount of unplanned purchases by more than 16%. Because the two strategies are not mutually exclusive, retailers could take advantage of both strategies simultaneously.

Finally, we report the results of a field experiment to directly assess the effectiveness of a mobile promotion strategy. Using shoppers' set of planned purchases, we provide a target coupon to each shopper for an unplanned category that either maximally increases his or her shopping path length or is adjacent to the planned path. The results of

our field experiment suggest that mobile promotion can significantly increase unplanned spending. Specifically, we find that a strategy that promotes an unplanned category that is farther from the planned purchase path substantially increased unplanned spending compared with a strategy that promotes an unplanned category near the planned purchases (\$21.29 vs. \$13.83).

Our research makes four important contributions to the shopper marketing literature. First, this is the first study to employ RFID tracking in conjunction with a shopping plan survey to understand the relationship between unplanned purchase behavior and in-store shopping path. Second, we derive a novel instrumental variable methodology that controls for endogeneity issues, allowing us to accurately estimate the direct effect of in-store travel distance on unplanned spending. Third, using our methodological framework and resulting estimates, we assess the relative effectiveness of mobile promotion strategies versus a product relocation strategy in increasing unplanned purchases. Finally, we conduct a field experiment that explores the effectiveness of mobile promotion and find that targeted mobile promotions aimed at increasing shopping path length are effective at increasing unplanned spending.

We organize the remainder of this article as follows: In the next section, we summarize the previous literature on the relationship between unplanned purchases and in-store shopping path and briefly discuss two shopper marketing strategies aimed at increasing in-store travel distances. In the following section, we construct our instruments and demonstrate their validity. We then provide an overview of a field study in which we estimate the relationship between in-store path length and unplanned spending and use our estimates in a simulation to assess the relative effectiveness of mobile promotion strategies compared with a benchmark, category relocation strategy. Finally, we report the results of a field experiment that directly tests the effect of mobile promotions on unplanned purchases and conclude with a discussion of managerial implications and future research directions.

Background and Literature Review

Relationship Between In-Store Travel Distance and Unplanned Spending

Previous research has revealed that shoppers often use physical products in a grocery store as external memory cues. Thus, exposure to in-store products and other stimuli often creates new needs, or reminds shoppers of temporarily forgotten needs, resulting in unplanned purchases (Kollet and Willett 1967). For example, based on a study of 11 consumers, Rook (1987, p. 193) reports that "the sudden urge to buy is likely to be triggered by a visual confrontation with a product or by some promotion stimulus." Because of the key role that exposure to physical products plays in driving unplanned spending, marketing researchers have long hypothesized that traveling further in a store will lead to more unplanned purchases (Granbois 1968).

Empirical evidence that directly measures the effect of in-store travel distance on unplanned spending, however, is

sparse, mainly because of the difficulty of measuring in-store path length accurately. Instead, researchers have typically relied on measures that are correlated with in-store travel distance such as number of store areas visited or number of aisles visited to provide corroborating statistical evidence. For example, Granbois (1968) employed research assistants, dressed as store employees, to discreetly follow shoppers as they moved around the store and record their in-store paths. By relating shoppers' in-store paths to their shopping baskets, he finds that shoppers who passed more locations within a store bought more product items. In a similar vein, Inman, Winer, and Ferraro (2009) use number of aisles visited as a proxy for the amount of exposure to in-store stimuli. They divide shoppers into three groups according to whether they visited "all aisles," "most aisles," or "a few aisles" and report that the group that visited "all aisles" has the highest likelihood of making unplanned purchases, followed by the group who visited "most aisles." None of the preceding work, however, accounts for the endogeneity of store path with regard to the extent of unplanned purchases.

Shopping time is another metric that researchers sometimes use as a proxy for in-store travel distance (and thus exposure to in-store stimuli). Granbois (1968) finds that shoppers who stay longer in the store are more likely to engage in unplanned purchases. For example, approximately 20% of those shoppers who spent five to six minutes shopping made one or more unplanned purchases, compared with only 8% of those shoppers who spent two minutes or less. Similarly, Inman, Winer, and Ferraro (2009) and Bell, Corsten, and Knox (2011) find a positive relationship between shopping time and unplanned spending. Other research shows that under time pressure, consumers exhibit less search activity (Beatty and Smith 1987) and make fewer unplanned purchases (Iyer 1989; Park, Iyer, and Smith 1989). We empirically demonstrate that in-store travel distance is a better measure of product exposure and more strongly related to unplanned spending.

Shopper Marketing Strategies Aimed at Increasing Travel Distance

Given that approximately half of shoppers' purchases are unplanned (Inman and Winer 1998; Inman, Winer, and Ferraro 2009; Point of Purchase Advertising Institute [POPAI] 1995), retailers and manufacturers are interested in shopper marketing strategies that facilitate unplanned purchases. Here, we review two main strategies: the "classic" strategy of managing product locations in the store, and the emerging technology of delivering promotions through location-based mobile shopping apps (hereinafter referred to as mobile promotion).

The product relocation strategy, first proposed by Granbois (1968) and reiterated in Iyer (1989), involves strategically placing popular product categories ("power categories" in Granbois 1968) in scattered locations throughout the store, which is similar to the conventional wisdom of "hiding the milk at the back of the store" among practitioners. The assumption is that by forcing shoppers to cover a longer distance in the store to find their planned

purchases, shoppers will be exposed to more in-store stimuli along the way and thus engage in more unplanned purchases. A rather extreme example is IKEA's "forced walk" layout, in which consumers are essentially forced to walk through the entire store from the entrance to the checkout and pass almost every product category (*The Scotsman* 2011). Another, less extreme, example is Hollister, in which the store layout is also dominated by one major pathway.

Recent advances in location-based mobile marketing present a new shopper marketing opportunity: offering targeted promotions aimed at increasing in-store travel distance and concomitant unplanned spending. For example, Foursquare recently announced a partnership with Safeway in which shoppers can link their loyalty card information to Foursquare and earn rewards for "checking in" (i.e., reporting that they just entered the store). Similarly, Shopkick automatically checks in the shopper, and then its partners (e.g., Best Buy) can offer in-store promotions. Relatedly, Modiv Media, whose clients include Giant and Stop & Shop (Zimmerman 2011), recently introduced a handheld scanner on which shoppers can scan their frequent shopper card and promotions are then sent to the device through a Wi-Fi network. Furthermore, as of June 2012, there were more than 500 iPhone apps (e.g., Grocery Gadget) that allow users to build a grocery shopping list using a prepopulated product list. Using the shopping list data, brand managers and retailers can then provide targeted coupon offers for unplanned categories to consumers via a mobile app (e.g., Ratz 2010).

To assess the effectiveness of both the classic strategy of product placements and the emerging strategy of mobile promotion, it is necessary to obtain an accurate estimate of the direct effect of in-store travel distance on unplanned spending. To that end, we go beyond providing corroborating statistical evidence, as previous researchers have done, by (1) measuring in-store travel distance accurately using RFID tracking and (2) addressing the endogeneity issues of in-store path, which we discuss in the next section.

Accounting for Endogeneity

In this section, we examine the endogeneity issue of in-store path length. We then develop and justify our instrumental variable approach to estimate the causal effect of in-store travel distance (PATHLEN) on the amount spent on unplanned purchases (UNPAMT).

Endogeneity of In-Store Travel Distance

As we mentioned previously, the endogeneity of in-store path length is the result of three factors: omitted in-store and out-of-store variables, simultaneity/reversed causality, and measurement error. First, omitted in-store and out-of-store variables can affect both unplanned purchases and in-store path lengths. Suppose that a shopper is attracted by a product display that promotes a certain product category, causing this shopper to deviate from his or her original path to approach and purchase that product. In such a case, both in-store path length and the amount of unplanned purchases increase, but there is no direct causal relationship between the two, resulting in a spurious correlation (Pearl 2000).

Another potentially omitted in-store variable is the social presence of other shoppers and the related crowding conditions (Argo, Dahl, and Manchanda 2005; Harrell, Hutt, and Anderson 1980; Hui, Bradlow, and Fader 2009). If an area is too crowded, a shopper may detour around the crowd (which increases in-store path length) and at the same time may become less likely to engage in a purchase (Harrell, Hutt, and Anderson 1980; Hui, Bradlow, and Fader 2009), again resulting in a spurious correlation between unplanned purchase and in-store path length (albeit in the opposite direction to the preceding example). In addition to in-store factors, omitted factors such as the shopper's impulsivity (e.g., Beatty and Ferrell 1998) may influence both unplanned purchasing and travel path in the store.

Second, because both in-store path length and the amount of unplanned purchases are generated during the same shopping trip, it is difficult to empirically tease apart the direction of causality between them. For example, it is possible that a shopper decides to buy an unplanned product first and then incurs the additional distance to purchase that product. Here, the decision to make unplanned purchases causes a longer in-store path rather than the other way around. More generally, shoppers may decide on their in-store path and unplanned purchases simultaneously, again causing in-store path length to be endogenous (Greene 2007).

Third, the in-store path length variable is invariably an imperfect measure. Measurement error is a serious limitation in earlier studies that employed research assistants to physically track shoppers' movements (Farley and Ring 1966; Granbois 1968; Underhill 2000) and remains an issue even in more recent studies that use RFID technology to track shoppers. The in-store paths collected by Hui, Bradlow, and Fader (2009), Hui, Fader, and Bradlow (2009b), and Larson, Bradlow, and Fader (2005) also have measurement errors because the RFID tags are attached to shopping carts rather than shoppers. That is, if a shopper leaves the cart at the end of an aisle before entering the aisle to shop, his or her movements will not be recorded. In our current study, this problem is mitigated because we put RFID tags on the *shopper*, though our measure is still noisy due to the error inherent in RFID signals. Measurement errors may cause endogeneity and lead to attenuation bias (e.g., Johnson and DiNardo 1997).

The Random Product Placement Model

We account for the endogeneity of in-store travel distance using an instrumental variable approach. Throughout this article, we use $i \in \{1, \dots, I\}$ to index shoppers. Thus, $UNPAMT_i$ denotes the dollar amount that shopper i spends on unplanned purchases, and $PATHLEN_i$ denotes the total distance, measured in feet, covered by shopper i in the store.

Following Gelman and Hill (2007), we construct and demonstrate the validity of our proposed instruments by appealing to an idealized thought experiment—the “random product placement” model:

1. Each shopper enters the store with a plan to buy a set of products. We assume that every shopper has the *same* plan.
2. Suppose that the product layout of the store randomly changes immediately before each shopper enters the store,

so that the locations of the planned products are randomly assigned for each shopper.¹

3. Assume that each shopper plans an in-store path (hereinafter referred to as “reference path”) to minimize the distance he or she must cover to pick up all planned purchases (based on store layout), by solving a traveling salesman problem (TSP) (Hui, Fader, and Bradlow 2009b). We denote the length of this path as $TSPLN$.
4. While in the store, the shopper may deviate from the reference path. For example, he or she may be attracted by some unplanned products in the store and thus incur additional travel distances. Thus, the shopper's actual in-store path is not exactly equal to the TSP path (i.e., $PATHLEN_i = TSPLN_i + \text{error}_i$).

Under the random product placement model, we cannot directly regress $UNPAMT$ on $PATHLEN$ to estimate the effect of travel distance, because $PATHLEN$ is clearly endogenous. However, in this setting, we can use $TSPLN$ as an instrument for $PATHLEN$. We check for the two conditions (relevance and exclusion) to determine the validity of $TSPLN$ as an instrument (Greene 2007). First, $TSPLN$ is clearly correlated with $PATHLEN$, and thus the relevance criterion is satisfied.² Second, because $TSPLN$ is randomly assigned through the random assignment of product placement, it cannot have any direct effect on unplanned purchases; thus, the “exclusive restriction” (Greene 2007) holds. Because $TSPLN$ is determined by only the store layout and computed *before* the shopper begins his or her trip, reversed causality or simultaneity can be ruled out through temporal precedence.

Although it has theoretical appeal as the most efficient path and is used in the previous literature as a frame of reference (Hui, Fader, and Bradlow 2009b), the TSP path is not the only reasonable candidate for a reference path. Specifically, rather than assuming that shoppers have the cognitive ability to solve the TSP, we can assume that shoppers only look forward one step (purchase) at a time. This is arguably a more realistic behavioral assumption considering that consumers are unlikely to devote significant cognitive resources to habitual tasks (Newell and Simon 1972). In addition, the ISLA assumption is more consistent with previous research in experimental economics (Camerer, Ho, and Chong 2004) and pedestrian modeling (Antonini, Bierlaire, and Weber 2006; Helbing and Molnar 1995). Here, the reference path is the path that connects the entrance, all the planned purchases by the order induced by a ISLA search algorithm, and the checkout (for details of the algorithm, see the Appendix). We denote the length of the reference path generated by the ISLA algorithm as $1SLALEN$.

TSPLN and 1SLALEN as Instruments with Observational Data

Subject to suitable controls, $TSPLN$ and $1SLALEN$ can both be used as valid instruments even with observational

¹We assume that the random assignment of product categories does not affect consumers' knowledge of the store layout.

²We verify this assumption in our empirical analysis. In our data set, the correlation between $TSPLN$ and $PATHLEN$ is .54 ($p < .001$); the correlation between $\log(TSPLN)$ and $\log(PATHLEN)$ is .45 ($p < .001$).

data. In contrast to the experimentally controlled setting in the random product placement model, the main feature of the actual observational setting is that each shopper comes into the store with a *different* set of planned items. Thus, the variation in TSPLN and ISLALN across shoppers is driven not by randomly assigned product locations (Assumption 2) but by the differences in shoppers' sets of planned categories. Thus, for TSPLN and ISLALN to still qualify as valid instruments, we must carefully control for other differences in the set of planned items that may have a direct effect on unplanned spending.

Recent research suggests that spending on unplanned purchases is driven by the amount of shoppers' "in-store slack" (Stilley, Inman, and Wakefield 2010a, b). Specifically, Stilley, Inman, and Wakefield (2010a) define in-store slack as the amount shoppers set aside for unplanned purchases in their mental trip budgets. Given the same shopping trip mental budget, a shopper with a greater number of planned purchases in mind will be less willing to purchase unplanned items because he or she does not have sufficient in-store slack remaining. For example, Stilley, Inman, and Wakefield show that there is a positive relationship between the number of unplanned purchases and the number of planned purchases, but this relationship reverses when shoppers' mental trip budget is included in the model. To control for this effect, which is directly caused by differences in the set of planned items, we introduce "mental budget slack," defined as the initial mental trip budget (measured using an entrance survey) minus the sum of average expenditures across shoppers on planned purchases, as a key control variable in our model.³

We now proceed to use ISLALN and TSPLN as instruments in our empirical analysis. We first provide an overview of our data, along with key summary statistics, and then discuss our estimation results using 2SLS (Greene 2007).

Study 1

Procedure

We conducted our first field study in a medium-sized grocery store located in a northwestern U.S. city. Figure 1 presents the layout of the store, which is divided into a total of 122 zones based on the locations of 99 product categories and discussion with store management. Table 1 presents the list of categories, their primary locations (in terms of store zone), and some category characteristics (refrigerated/non-refrigerated and category size).⁴

We collected data from 300 shoppers. Of these, 25 participants provided incomplete responses to either the entrance or exit survey, leaving 275 shoppers for analysis. Upon arriving at the store through its only entrance (lower

right-hand corner in Figure 1), shoppers were approached and invited to participate in a marketing research study. Each participant then completed an entrance survey. The questions included (1) whether they had a shopping list today, (2) their expected expenditure for the trip (i.e., their mental trip budget), (3) whether they were shopping alone, and (4) their familiarity with the store in terms of the product locations. Finally, on a list of all 99 product categories in the store (see Table 1), they checked all the products they planned to purchase during the current shopping trip. This approach lessens the likelihood of overestimating unplanned spending due to forgetting or fatigue on the entrance survey. In exchange for participating, shoppers received a \$5 store gift card, which we gave them after they finished shopping to avoid a windfall effect.

Table 1 also presents the percentage of shoppers who planned to buy in each category. To make sure that asking the mental budget question beforehand did not have an influence on shopping intentions, we reversed the order of the mental budget and shopping intentions questions for half the participants. Consistent with Stilley, Inman, and Wakefield (2010a), we found no effect of question order. Furthermore, consistent with Stilley, Inman, and Wakefield (2010a, b), the correlation between the stated mental budget and actual spending is strongly positive ($r = .826, p < .001$).⁵

After finishing the entrance survey, the experimenter helped participants don a PathTracker belt, embedded with an RFID tag, developed by TNS Sorensen to track shopper movements (see Hui, Bradlow, and Fader 2009; Hui, Fader, and Bradlow 2009b). The RFID tag on the PathTracker belt emits a radio frequency signal every five seconds, which is then picked up by the antennas at the perimeter of the store, allowing us to track the (x, y) coordinate of the shopper in the store. We compute the total in-store path length (PATHLEN) for each shopper from his or her shopping path. Figure 2 presents several example shopping paths from our data set. As the figure shows, similar to previous findings (Hui, Bradlow, and Fader 2009; Hui, Fader, and Bradlow 2009b; Larson, Bradlow, and Fader 2005), shoppers rarely cover the entire store during their shopping trips. In our data set, we find that in terms of the 122 store zones (shown in Figure 1), on average, shoppers cover only approximately 37% of the store, with a minimum of 7% and a maximum of 72%.

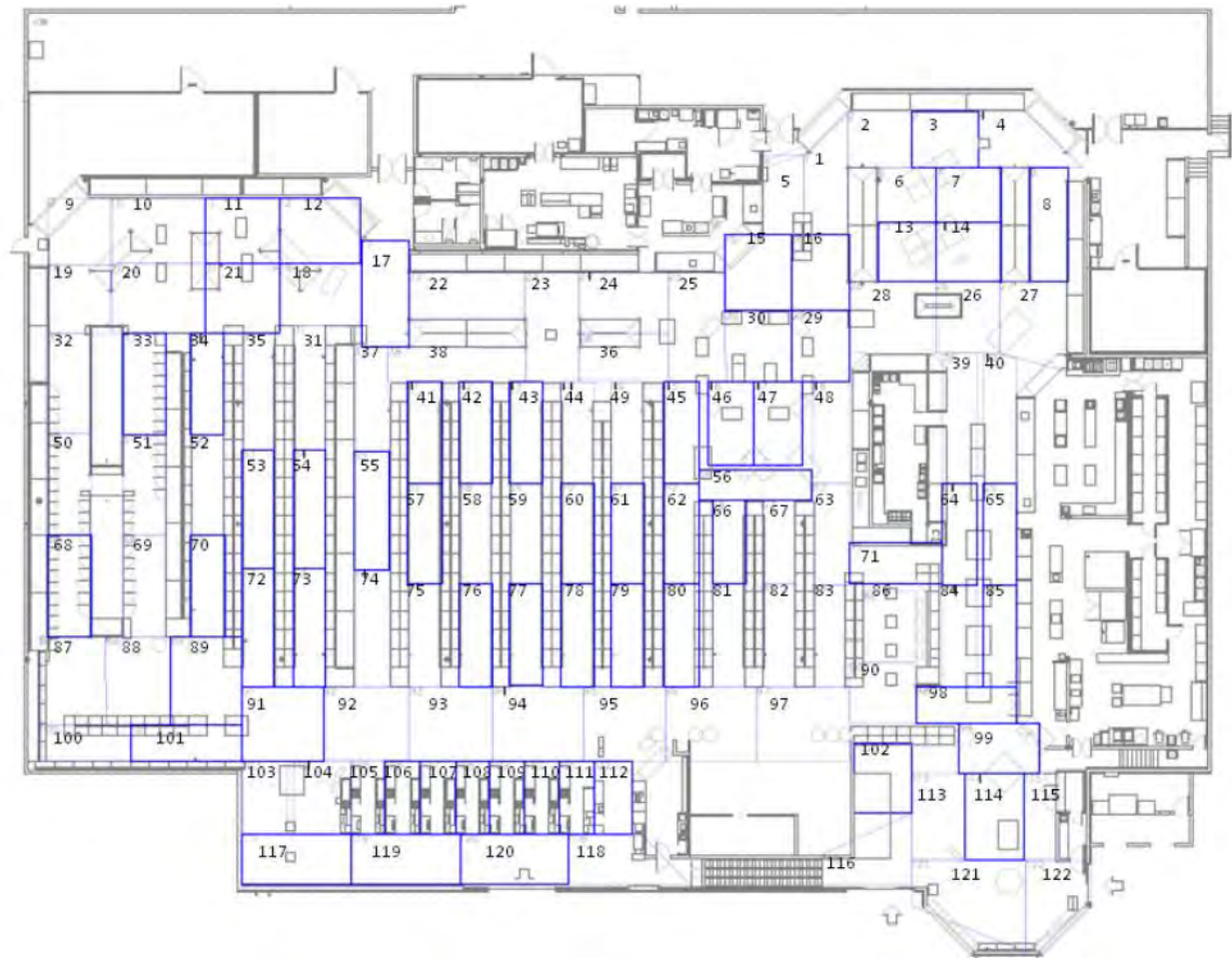
After finishing their shopping trip and checking out, participants completed an exit survey in which they answered several demographic questions, including their gender, age, household size, household income, whether they have children, and distance from the store. Following Stilley, Inman, and Wakefield (2010b), we also assessed each shopper's impulsivity trait using Rook and Fisher's (1995) nine five-point semantic differential scales. Finally, each shopper was given a \$5 gift card, thanked, and then

³We also tested an alternative specification in which the in-store slack variable is defined as the total mental budget minus the total amount that each shopper spent on his or her own planned purchases. The results are similar to those presented here.

⁴Some product categories are located in more than one store zone. In such cases, we use the location that has the highest density of stockkeeping units related to that product category.

⁵Stilley, Inman, and Wakefield (2010a) assess whether eliciting shoppers' mental budget and planned items influences the amount spent. They use the retailer's frequent shopper program data to compare the amount spent on the survey date with the shoppers' average amount spent on similar trips in the prior six months. They find no difference (\$58.42 vs. \$59.16).

FIGURE 1
Study 1: Store Layout Divided into 122 Zones



dismissed. The store provided the transaction history for each shopping trip for all participants. By subtracting the amount that each shopper spent on planned categories stated in his or her entrance survey from total spending, we computed the amount of money that each shopper spent on unplanned purchases (UNPAMT).

Table 2 reports summary statistics for the amount spent on unplanned purchases, together with in-store path length and other shopper demographic variables. On average, shoppers traveled approximately 1400 feet in the store (approximately one-third of a mile), and the average amount spent on unplanned purchases was approximately \$16, roughly 40% of their total shopping budget. This is consistent with the findings Stilley, Inman, and Wakefield (2010a, b) report.

Following the procedures outlined in the Appendix, we computed the length of the reference path generated by TSPLN and 1SLALEN for each shopper, using the locations of their planned categories. (Summaries for TSPLN and 1SLALEN are listed under “Instrumental Variables” in Table 2.) The average TSPLN was 562 feet, while the average 1SLALEN was 607 feet. This suggests that only looking one step ahead rather than considering their entire

set of planned purchases results in shoppers traveling 8% farther. On average, the actual in-store path length shoppers travel is 2.5 times longer than TSPLN and 2.3 times longer than the 1SLALEN. The correlation between $\log(\text{PATHLEN})$ and $\log(\text{TSPLN})$ is .45, the correlation between $\log(\text{PATHLEN})$ and $\log(1\text{SLALEN})$ is .46, and the correlation between $\log(\text{TSPLN})$ and $\log(1\text{SLALEN})$ is .98. Because of the high correlation between the two variables and because 1SLALEN is a more realistic representation of shoppers’ optimization abilities given their cognitive motivation and processing constraints, we only use 1SLALEN as an instrument in our subsequent analyses.⁶

Results

We apply log-transformations to both the dependent variable UNPAMT and the independent variable PATHLEN. Conceptually, after a shopper has traveled a long distance in the store, any additional travel distance will likely lead to a smaller increase in unplanned spending, so measuring both variables in log-scale (percentage terms) seems appropriate. Statistically, we find that the log-transform to both PATHLEN

⁶The results with TSPAEN as the instrument are almost identical.

TABLE 1
List of Product Categories (with Store Location), Sorted by Penetration

Category Name	Zone	%Buy	%Plan	Category Name	Zone	%Buy	%Plan
Fresh fruit ^{NL}	28	65.3%	38.0%	Wraps/plastic/foil/bags ^{NM}	62	3.3%	2.7%
Fresh vegetables/potatoes ^{NL}	4	46.0%	40.3%	Health/beauty (others) ^{NL}	31	3.3%	1.0%
Bread/rolls/buns ^{NL}	81	31.7%	27.3%	Canned fruits ^{NM}	60	3.0%	2.7%
Yogurt ^{RM}	9	30.7%	13.0%	Cookies ^{NL}	52	3.0%	3.7%
Milk ^{RS}	10	28.3%	45.3%	Nuts ^{NL}	74	3.0%	2.0%
Cheese ^{RM}	11	26.3%	14.7%	Dish soap/detergent ^{NS}	75	3.0%	2.3%
Fresh baked goods ^{NM}	64	23.3%	14.7%	Ice ^{RS}	117	3.0%	3.3%
Prepared meats ^{RL}	12	22.3%	9.7%	Pet supplies ^{NM}	35	2.7%	3.0%
Fresh meat ^{RM}	38	22.0%	23.0%	Pudding/Jell-ONS	42	2.0%	.7%
Fruit juices/drinks ^{NL}	74	19.3%	15.3%	Peanut butter/jam ^{NS}	66	2.0%	1.3%
Eggs ^{RS}	20	18.7%	19.0%	Frozen bread/pies ^{RM}	68	2.0%	1.0%
Salty snacks ^{NL}	55	17.0%	7.3%	Frozen meat/seafood ^{RM}	69	2.0%	2.3%
Wine ^{NL}	46	15.3%	11.3%	Creams/lotions ^{NM}	54	2.0%	.3%
Frozen ice cream ^{RM}	50	15.3%	16.0%	Tobacco ^{NM}	95	2.0%	1.3%
Soft drinks ^{NL}	100	14.7%	14.7%	Snack bars ^{NL}	70	1.7%	2.3%
Cereal ^{NM}	52	12.3%	11.0%	Snacks (others) ^{NM}	5	1.7%	2.3%
Fresh fish ^{RM}	15	12.0%	11.0%	Frozen breakfasts ^{RM}	50	1.7%	1.7%
Coffee/tea ^{NL}	59	11.7%	4.3%	Frozen potato ^{RS}	33	1.7%	.7%
Cream ^{RS}	10	11.7%	7.0%	Paper towels ^{NS}	54	1.7%	2.0%
Canned vegetables ^{NM}	78	10.7%	9.7%	Oral care ^{NM}	54	1.7%	1.3%
Salad dressing ^{NM}	77	10.7%	10.7%	Magazines ^{NM}	106	1.7%	1.0%
Condiments ^{NL}	59	10.3%	7.0%	Vinegar ^{NM}	14	1.3%	1.0%
Beer ^{RM}	37	9.7%	9.7%	Dough products ^{NS}	32	1.0%	1.3%
Precut fresh salad mix ^{RM}	27	9.3%	9.7%	Frosting ^{NM}	42	1.0%	.3%
Rice/beans/pasta ^{NL}	61	9.3%	6.0%	Bath soap ^{NS}	57	1.0%	1.3%
Frozen dinners ^{RM}	51	9.0%	4.0%	Deodorant ^{NS}	54	1.0%	.3%
Margarine/butter ^{RS}	32	8.7%	4.7%	Pharmacy (others) ^{NL}	73	1.0%	1.0%
Crackers ^{NL}	70	8.3%	7.3%	Spices/seasoning ^{NL}	76	.7%	4.0%
Fresh poultry ^{RM}	24	8.0%	6.7%	Frozen snacks ^{RM}	51	.7%	.0%
Bottled water ^{NL}	100	6.7%	5.3%	Dried fruit ^{NM}	5	.7%	.3%
Candy ^{NL}	81	6.7%	2.3%	Marshmallows ^{NS}	42	.7%	.7%
Canned soup ^{NM}	62	6.3%	4.3%	Batteries ^{NM}	112	.7%	1.0%
Cake mix ^{NM}	42	5.7%	4.7%	Pain relief/gastrointestinal ^{NS}	54	.7%	.0%
Household cleaners ^{NM}	57	5.7%	3.0%	Baby ^{NM}	67	.3%	.7%
Mexican sauce ^{NM}	49	5.0%	2.7%	Chewing gum ^{NS}	108	.3%	1.3%
Pasta sauce ^{NL}	79	5.0%	1.7%	Frozen pizza ^{RS}	69	.3%	.7%
Baking soda ^{NM}	58	4.3%	2.7%	Feminine hygiene ^{NS}	67	.3%	.7%
Sugar/syrup ^{NS}	42	4.3%	4.7%	Shampoo/conditioner ^{NS}	54	.3%	.7%
Sports/energy drinks ^{NL}	88	4.3%	5.7%	Nutritional supplements ^{NS}	54	.3%	.3%
Shelf stable meals ^{NM}	61	4.3%	2.3%	School supplies ^{NM}	82	.3%	.7%
Cooking oil ^{NM}	76	4.0%	2.7%	Fruit rolls ^{NM}	70	.0%	.3%
Toilet paper ^{NM}	31	4.0%	4.7%	Popcorn ^{NS}	55	.0%	2.0%
Pickles/olives ^{NM}	43	3.7%	3.0%	Toaster pastries ^{NS}	70	.0%	1.0%
Floral ^{NM}	115	3.7%	2.0%	Diet food ^{NS}	78	.0%	.3%
Greeting cards ^{NL}	83	3.7%	2.7%	Side dishes ^{NM}	19	.0%	2.7%
Canned meat/fish ^{NM}	49	3.3%	3.0%	Candles ^{NS}	42	.0%	.3%
Frozen vegetables ^{RM}	33	3.3%	3.3%	Facial tissue/napkins ^{NM}	54	.0%	.7%
Hardware ^{NM}	72	3.3%	.0%	Mixers ^{NS}	72	.0%	.3%
Kitchenware ^{NL}	86	3.3%	.3%	Picnic supplies ^{NM}	80	.0%	.7%
Laundry ^{NM}	41	3.3%	4.0%				

Notes: R = refrigerated, and N = nonrefrigerated; S = small, M = medium, and L = large.

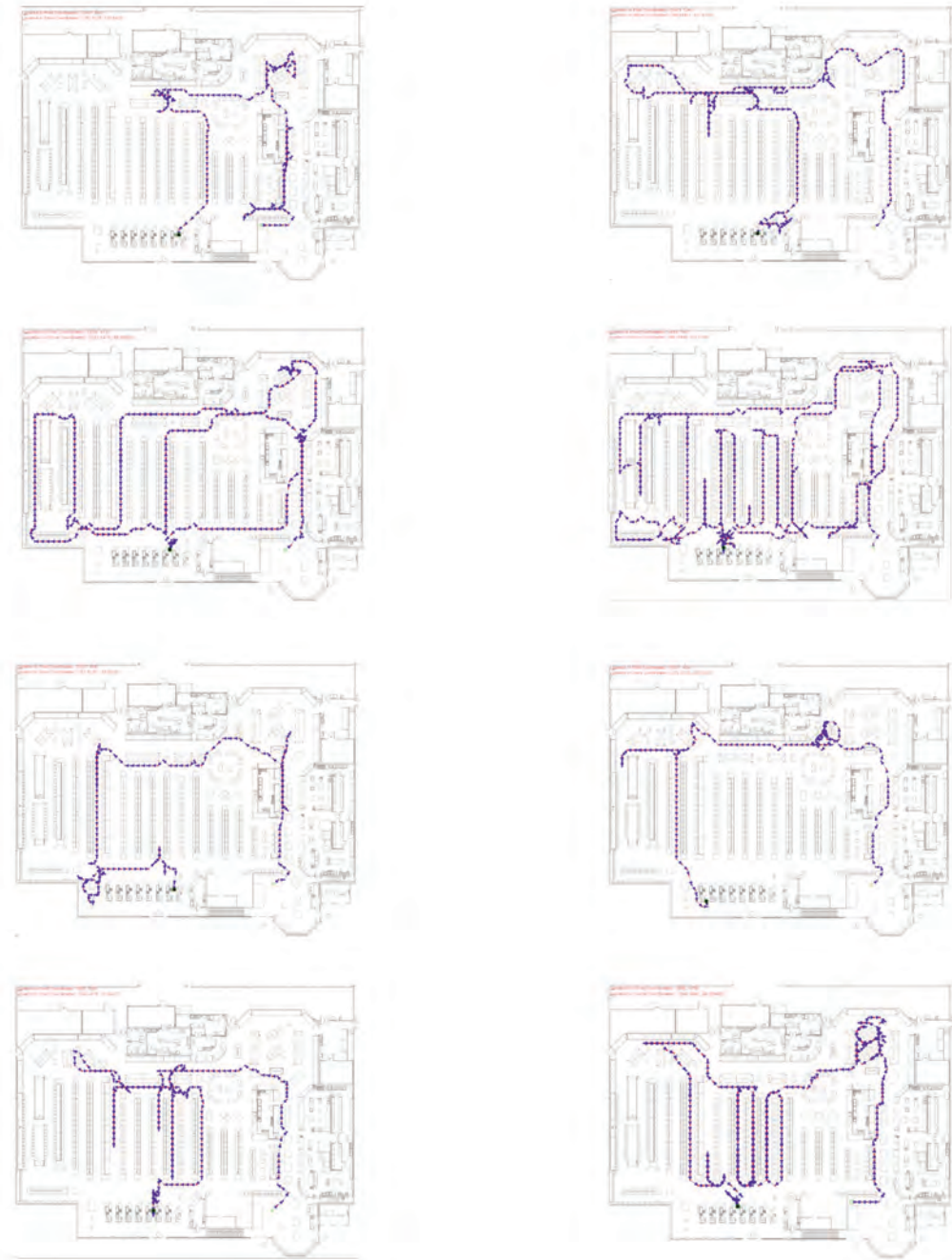
and UNPAMT stabilizes the variances across observations and helps avoid the problem of heteroskedasticity.⁷ Thus, we estimate the coefficient of log(PATHLEN) on log(UNPAMT + 1) by first performing OLS regression with

⁷To check for the robustness of our empirical results, we also conducted the analysis on the untransformed variables. The results are substantively unchanged: the 2SLS estimates (with either TSPLN or 1SLALEN as the instrument) for the coefficient of PATHLEN are approximately 40% greater than the uncorrected OLS estimates.

all control variables, but without including any instruments, and then performing 2SLS with log(1SLALEN) as an instrument. We estimated the OLS and 2SLS regression using the lm and tsls packages in R, respectively. Table 3 presents the estimated coefficients.

As Table 3 indicates, across both specifications, the coefficient of log(PATHLEN) is positive and highly significant ($p < .001$). However, the magnitude of the coefficient differs widely between the two models: The coefficient in the model with log(1SLALEN) as the instrument is 57% higher

FIGURE 2
Example Shopping Paths Collected Using RFID in Study 1



than the OLS estimate. To confirm that $\log(\text{PATHLEN})$ is endogenous, we conduct a Hausman specification test (Hausman 1978). A small p -value ($<.05$) of the Hausman test indicates that the 2SLS estimate is significantly different from the uncorrected OLS estimate, indicating endogeneity in the regressor. The Hausman statistics for the model with $\log(\text{ISLALLEN})$ as instrument is $H = 6.18$ ($p = .013$), confirming the endogeneity bias of the uncorrected OLS estimate. In addition, we note that the first-stage R -square value is .292, and the first-stage partial F -statistic in the first-stage regression is 48.33 ($p < .001$), which is well

above the $F > 10$ criterion for strong instruments (Staiger and Stock 1997; Stock, Wright, and Yogo 2002).

On the basis of the 2SLS results, we conclude that the elasticity of the amount spent on unplanned purchases on travel distance is 1.57. To put this effect in monetary terms, a shopper marketing strategy that increases path length by 10% for each shopper (i.e., an average of 140 feet) will increase unplanned spending by approximately 16.1%, or \$2.54 per shopper.

Next, we turn to the coefficients on the set of control variables in our study. As expected, in-store slack has a

TABLE 2
Key Summary Statistics of Study 1 Data Set

Variable	Description	M	SD	Mdn	Min	Max
Dependent Variable						
UNPAMT	\$ spent on unplanned purchases	15.79	16.60	10.74	.00	88.20
Independent Variables						
PATHLEN	Total in-store path length (in feet)	1398.85	708.38	1258.40	43.90	4098.00
IMP	Impulsivity (Rook and Fisher 1995)	2.32	.74	2.30	1.00	4.56
BUDGET	Shopping budget (\$)	39.47	33.87	30.00	2.00	300.00
SLACK	In-store slack (Budget–planned expenditure)	9.74	23.57	4.84	–40.95	131.70
GENDER	Gender (1 = male)	34%	—	.00	.00	1.00
HSIZE	Household size (1 = single; 0 = two or more)	17%	—	.00	.00	.00
CHILDREN	Has children (1 = yes)	25%	—	.00	.00	1.00
LIST	Shopping list (1 = yes)	36%	—	.00	.00	1.00
DIST	Distance from store (miles)	6.11	18.15	3.00	.10	250.00
ALONE	Shopping alone (1 = yes)	81%	—	1.00	.00	1.00
KNOW	Knowledge of store (1–5)	4.15	1.03	4.00	1.00	5.00
AGE	Age (1 = 45 years or older)	76%	—	1.00	.00	1.00
INCOME	Income (1 = > \$75K; 0 = < \$75K or no response)	44%	—	.00	.00	1.00
Instrumental Variables						
TSPLEN	Length of the reference path generated using TSP algorithm	562.44	158.02	559.38	76.01	1034.69
1SLALEN	Length of the reference path generated using 1SLA algorithm	606.87	193.97	586.44	76.01	127.74

TABLE 3
Coefficient Estimates with OLS and 2SLS with log(1SLALEN) as Instrument

Variable	OLS Estimates		2SLS Estimates (1SLALEN)	
	Estimate	SE	Estimate	SE
Intercept	–5.441**	.802	–9.416**	1.793
log(PATHLEN)	.999**	.102	1.573**	.252
SLACK	.009**	.003	.006*	.003
IMP	.234**	.079	.179**	.087
GENDER	–.017	.123	.083	.136
AGE (45 years and older)	.147	.142	.058	.154
INCOME (>\$75k)	.217*	.118	.240*	.126
HSIZE	–.009	.159	.002	.168
CHILDREN	.368**	.138	.405**	.147
LIST	–.070	.122	–.184	.137
DIST	.000	.003	.001	.003
ALONE	–.158	.147	–.083	.158
KNOW	–.034	.058	–.024	.061
R ²	.415		.344	

* $p < .10$.

** $p < .05$.

positive effect on the amount of unplanned purchases ($p < .10$). This is consistent with Stilley, Inman, and Wakefield (2010a) and reflects the purpose of in-store slack to fund unplanned purchases. Across both regressions, higher impulsivity is, as expected, positively related to the amount of unplanned spending ($p < .05$). Moreover, higher income and having children are positively related to higher amount spent on unplanned purchases. This is similar to Bell, Corsten, and Knox (2011) and Inman, Winer, and Ferraro (2009), providing face validity for our model estimates. The positive income coefficient suggests that higher-income shoppers have greater discretion to make additional unplanned purchases, while the positive coefficient for the presence of children may reflect either the greater likeli-

hood of in-store need recognition or less time available to adequately plan.

Robustness Checks

To ensure that our results are robust to different model specifications, we performed three additional analyses that (1) explore the role of shopping time, (2) control for the number of planned categories, and (3) study the number of unplanned categories as a dependent variable. Tables 4 and 5 summarize the results.

First, as we mentioned in the literature review, previous research has studied the relationship between shopping time and the extent of unplanned purchases. Because traveling longer in the store requires staying longer in the store, shop-

TABLE 4
Results of Robustness Checks

Variable	Robustness Check with Number of Planned Categories		Robustness Check with Large/Small Planned Basket as Binary Covariate		Robustness Check with Shopping Time Instead of Distance	
	Estimate	SE	Estimate	SE	Estimate	SE
Intercept	-1.002*	5.296	-8.407**	3.082	-2.286**	.769
log(PATHLEN)	1.665**	.794	1.421**	.451	—	—
log(Shoptime)	—	—	—	—	1.450**	.247
NPlanCat	-.009	.059	—	—	—	—
Large	—	—	.129	.247	—	—
SLACK	-.005	.004	.006*	.003	.006*	.003
IMP	.178*	.091	.182**	.085	.194**	.091
GENDER	.095	.170	.075	.136	.190	.152
AGE (45+)	.045	.189	.083	.162	.063	.164
INCOME (>\$75K)	.247*	.138	.228*	.126	.300**	.135
HSIZE	.005	.173	-.016	.168	-.104	.179
CHILDREN	.414**	.165	.389**	.148	.460**	.157
LIST	-.188	.146	-.184	.134	-.171	.145
DIST	.001	.003	.001	.003	-.002	.003
ALONE	-.079	.167	-.098	.159	.052	.175
KNOW	-.024	.063	-.021	.060	.002	.066
R ²		.320		.379		.260

* $p < .10$.

** $p < .05$.

Notes: We included the number of categories in the planned basket as a continuous covariate (NPlanCat); we included a binary variable indicating large or small planned basket as a covariate (Large); and we used shopping time (log(Shoptime)) rather than in-store travel distance. We estimated all models using 2SLS.

TABLE 5
**Coefficient Estimates of a Poisson Regression
with the Number of Unplanned Categories
Purchased as Dependent Variable and
log(1SLALEN) as Instrument**

	Estimate	SE
Intercept	-6.771**	1.145
log(PATHLEN)	1.097**	.162
SLACK	.001	.002
IMP	.139**	.048
GENDER	-.041	.082
AGE (45 years and older)	-.089	.087
INCOME (>\$75K)	.141*	.072
HSIZE	.043	.103
CHILDREN	.269**	.079
LIST	-.123	.078
DIST	-.001	.002
ALONE	-.019	.084
KNOW	-.055*	.034

* $p < .10$.

** $p < .05$.

ping time and in-store travel distance are positively correlated ($r = .52, p < .001$). However, we argue that, conceptually, in-store travel distance is a better proxy for the amount of exposure to in-store product stimuli than shopping time. On the one hand, if a shopper spends a long time in the store by lingering at a few locations, exposure to in-store product stimuli will be limited even if shopping time is long. On the other hand, in-store travel distance is directly related to the additional products that the shopper will pass by during the shopper's trip, which is particularly relevant considering that shoppers in our sample on average only

covered 37% of the store. Thus, we expect that compared with shopping time, in-store travel distance should be more strongly related to unplanned spending.

This logic is supported by our empirical analysis. An OLS regression with shopping time instead of in-store travel distance (along with all the other control variables listed in Table 2) results in a R-square value of .382, which is lower than the R-square value of .415 with in-store travel distance (see Table 3). If *both* shopping time and in-store travel distance are introduced into the regression, the resulting R-square value only becomes slightly higher (.417). Furthermore, while in-store travel distance is highly significant ($p < .001$), shopping time is insignificant ($p = .30$). We obtained similar results when we estimated the regressions using 2SLS with log(1SLALEN) as an instrument. The second stage R-square value for the model with shopping time instead of in-store travel distance is .260 (as shown in the third column in Table 4), which is much lower than the second-stage R-square value of .344 (see Table 3) for the model with in-store travel distance. Taken together, these analyses suggest that in-store travel distance is more strongly related than shopping time to unplanned spending.

Next, we conducted a set of additional analyses to ensure that our results were not driven by differences in basket sizes across shoppers. Note that as explained previously, we already controlled for differences in basket sizes through the in-store slack variable; that is, given the same mental budget, a shopper with more planned categories will have less in-store slack to spend on unplanned purchases. To further ensure that we adequately controlled for the size of the planned set, we conducted two additional analyses. The first includes the number of planned categories as an

additional control variable, and the other includes a binary variable indicating large/small planned list based on a median split.

Importantly, in both analyses, the estimated coefficients (using 2SLS) for $\log(\text{PATHLEN})$ are similar to the coefficient (1.57) presented in Table 3. The estimated coefficient of $\log(\text{PATHLEN})$ is 1.67 when the number of planned categories is included as a control variable (see the first column of Table 4) and 1.42 when a large/small planned list binary variable is included (see the second column of Table 4). In both cases, the planned basket size variable is not significant ($p = .88$ and $.60$ for the first and second analyses, respectively). This suggests that the size of the planned basket is already adequately controlled for by the in-store slack variable.

Finally, in the previously mentioned results, we used the amount of unplanned spending as the dependent variable. To further explore the robustness of our results, we conducted an analysis with the number of unplanned categories as the dependent variable, a measure that is common in studies of unplanned purchases (e.g., Bell, Corsten, and Knox 2011). This mitigates the concern that the variability in prices paid for different categories are distorting our results. Specifically, we estimate a Poisson regression using the number of unplanned categories purchased as a dependent variable and $\log(\text{ISLALLEN})$ as an instrument. Table 5 shows the estimated coefficients. As the table indicates, the results are similar to those in Table 3. We again found a positive coefficient for in-store path length, albeit smaller in magnitude ($\beta = 1.10, p < .001$). Similar to Table 3, we found that a positive coefficient for impulsivity ($\beta = .139, p < .01$) and high income ($\beta = .141, p < .10$). In addition, we found a positive effect for children ($\beta = .269, p < .001$) and a negative effect for store knowledge ($\beta = -.055, p < .10$).

Assessing Shopper Marketing Strategies to Increase Path Length

Using our methodological framework and parameter estimates, we conduct two simulations to study the relative effectiveness of the emerging strategy of mobile promotion on increasing unplanned spending versus the product relocation strategy originally proposed by Granbois (1968). Our simulation procedure is grounded in the paradigm of agent-based models (Rand and Rust 2011), which are widely used to model pedestrian behavior in urban studies (Batty 2001) and product diffusion in marketing (Goldenberg et al. 2007; Goldenberg, Libai, and Muller 2010). In agent-based models, researchers begin with a set of simple yet reasonable assumptions and rules on how agents (shoppers) interact with their environment (plan their shopping trips). Then, interventions are made to the environment, and researchers predict how agents will behave under the new setting using simulations. Here, we explicitly begin with the assumption that shoppers plan their trip using a reference path produced by the ISLA algorithm and that their actual in-store path length may deviate from the reference path (see assumption 4). We then hypothetically change the retail environment by offering targeted promotions to attract shoppers and simulate how shoppers' unplanned spending will change under

the new environment. We benchmark the results of this analysis against a product relocation strategy.

Each simulation involves the following steps. We first recompute $\log(\text{ISLALLEN})$ for each shopper under the considered strategy (i.e., switching the position of product categories, or adding a new product category into a shopper's planned set through in-store targeted promotions). Then, we use the change in $\log(\text{ISLALLEN})$ to compute the resulting change in $\log(\text{PATHLEN})$ using the coefficient of $\log(\text{ISLALLEN})$ in the first-stage regression (.673). Finally, we use the predicted change in $\log(\text{PATHLEN})$ to compute the change in unplanned spending using the coefficient of $\log(\text{PATHLEN})$ in the second-stage regression (1.573).

Benchmark strategy: product category relocation. We consider strategically switching the locations of up to three pairs of popular product categories. We deliberately focus on small-scale changes of product placements because it is more realistic from the retailer's perspective. Moreover, we find that the incremental effect of moving more product categories levels off after three category pairs (details are available upon request). Following Granbois (1968), we consider the top 20 "power" product categories consumers most frequently plan, as Table 1 indicates. To take the physical constraints of product categories into account, we consider only switches between product categories that are of the same "type," in terms of the need for refrigeration (refrigerated/nonrefrigerated) and size (small/medium/large), both indicated by the superscripts in Table 1.⁸ After taking these constraints into account, there are a total of 43 feasible pairwise switches among the top 20 "power" categories, resulting in 12,341 feasible combinations of pairwise switches when three product category pairs are considered.

The combination of pairwise category switches that jointly results in the largest increase in unplanned spending is (1) fresh fruit (zone 28) with fresh vegetable/potatoes (zone 4), (2) fresh baked goods (zone 64) with cereal (zone 52), and (3) fresh meat (zone 38) with precut fresh salad mix (zone 27).⁹ Under these recommendations, shoppers' reference paths are expected to increase in length by approximately 5.5%. As a result, unplanned spending is predicted to increase by 7.2%, or an average of \$1.14 per shopper. Importantly, this result suggests that the effect of product relocation is not particularly dramatic, presumably because the retailer has already fine-tuned store's layout due to various adjustments and experimentations over time.

Mobile promotions. Location-based mobile apps can be used to deliver in-store targeted promotions to increase shoppers' in-store travel distances and thus unplanned spending. Specifically, shoppers check in at a store when they enter it (e.g., using Foursquare) and enter a list of product categories that they plan to buy using a grocery app (e.g., Grocery IQ) that partners with the retailer. Because mobile

⁸We also conducted a sensitivity analysis by considering an alternative specification in which no constraint is imposed. The results are similar to the ones reported here and are available on request.

⁹For succinctness, we only briefly report the simulation results here. Details are available on request.

GPS may not be accurate enough to locate the shopper's precise in-store location, we do not use a shopper's within-store location for the purpose of targeted promotions.

For our simulations, we assume the following mechanism for delivering in-store targeted promotions to shoppers. As we stated previously, a shopper enters her list of planned purchases in the grocery app.¹⁰ Then, using the planned shopping list, the grocery app provides three targeted offers (the same number of offers in Danaher et al. 2011) via the shopper's smartphone, with the goal of maximally increasing the length of the shopper's reference path by adding new product categories into his or her planned list if the offer is redeemed. In our simulations, we specify a promotion response rate of 20%, the industry average reported by Moto Message (www.motomessage.com) and also consistent with Danaher et al. (2011).

For each shopper, we consider all possible combinations of three-category offers. For each combination, we compute the expected change in $\log(1SLALEN)$ given that each offer has 20% probability of being redeemed. More specifically, if an offer is redeemed, the corresponding product category will be added to the shopper's planned list and thus increase the length of the reference path.¹¹ Next, we select the combination of offers that provides the highest expected increase in $\log(1SLALEN)$. Then we compute the corresponding predicted change in $\log(PATHLEN)$ and thus the resulting increase of unplanned spending using the estimated elasticity.

Table 6 presents the top five categories most frequently promoted, along with the predicted effectiveness of the mobile promotion strategy. As the table shows, the mobile promotion results are promising. The predicted increase in

$\log(1SLALEN)$ is .149, which leads to a predicted increase of 16.1% in unplanned spending, or \$2.54 per shopper. This compares favorably with the effectiveness of the benchmark product relocation strategy results reported previously, which generate a predicted 7.2% lift in unplanned spending.

The main reason that targeted mobile promotion can be effective is that, unlike product category relocations, promotions can be customized for each shopper according to his or her prospective path. Thus, retailers can take into account the heterogeneity in shoppers' plans to target promotions and increase their travel distance strategically. Conversely, product relocation affects all shoppers, limiting the amount of control the retailer can exercise. Table 6 reports some evidence supporting this claim, showing the five product categories that should be most frequently promoted. The most frequently promoted categories are fresh vegetable/potatoes (located in the upper right-hand corner of the store), yogurt (located in the upper left-hand corner of the store), ice and frozen bread/pies (located in the lower left-hand corner of the store), and shelf stable meals (located in the middle of the center-of-store aisles). This shows the unique ability of in-store *targeted* promotions to drive in-store path patterns. That is, if a shopper's original planned path does not include a visit to some of the store corners or the middle of the store, retailers can encourage the shopper to cover more of the store by promoting the categories located there. In contrast, this cannot be done by product relocation, because a product relocation affects all shoppers. For this reason, targeted promotions through location-based mobile apps can be more effective and probably less costly than product relocations in increasing in-store travel distance and, concomitantly, unplanned spending.

Study 2

Procedure

To further explore the effectiveness of mobile promotion, we conducted a controlled field experiment in a grocery store in Pittsburgh. This store is much larger than the one we used in Study 1 to estimate our model, which increases the generalizability of our findings. Because mobile promotion technology is in its nascent stage, we sought to simulate the process discussed previously, through which a coupon for an unplanned category would be delivered to shoppers early in their trip. To do this, we assessed shop-

TABLE 6
Results of the Simulation of In-Store Targeted Promotions

Category	Zone	Promotion Frequency
Fresh vegetable/potatoes	4 (upper right-hand corner)	45.1%
Yogurt	9 (upper left-hand corner)	15.6%
Ice	117 (lower left-hand corner)	14.9%
Frozen bread/pies	68 (lower left-hand corner)	12.4%
Shelf-stable meals	61 (center of store)	11.3%
Average change in $\log(1SLALEN)$ = .149		
Average \$ change in unplanned spending = \$2.54		
Average % change in unplanned spending = 16.1%		

¹⁰Note that we assume that shoppers honestly reveal their shopping lists. Because shoppers have no reason to expect to get coupons on unplanned items, they have little incentive to strategically hide their shopping lists. To further discourage shoppers from "gaming the system," a potential alternative would be to offer promotions for both on- and off-the-list items, so that consumers cannot easily figure out the targeting strategy.

¹¹Computationally, eight scenarios are possible: none of the offers is redeemed (with probability $.8 \times .8 \times .8 = .512$), one of three offers is redeemed (three cases, each with probability $.2 \times .8 \times .8 = .128$), two of three offers are redeemed (three cases, each with probability $.2 \times .2 \times .8 = .032$), or all three offers are redeemed (with probability $.2 \times .2 \times .2 = .008$). We compute the expected change in $\log(1SLALEN)$ by multiplying the probability of each scenario with its effect on $\log(1SLALEN)$ and then summing across scenarios.

pers' planned items at the point of entry to the store and then used a laptop to enter their planned purchases into a computer program (written in R) that compared the planned categories against a store map and generated a coupon according to the experimental condition. The coupons were selected from a set of popular categories at various locations in the store: paper products (toilet/facial tissue and paper towels), canned soup, oral care, milk and eggs, over-the-counter medicines, cereal, bottled water, flour/cake mix, cookies and crackers, pasta, and ice cream.

The study focus is to evaluate the effect on unplanned spending of using in-store coupons to incentivize shoppers to visit more of the store. Thus, we employed a 2 (couponed category proximity to planned path: near vs. far) $\times$ 2 (coupon amount: \$1 vs. \$2) design. Shoppers were randomly assigned to one of the four experimental conditions with equal probability. Shoppers in the far condition received a coupon for an unplanned category that was relatively far from the shopper's planned set (i.e., that would result in the largest increase in ISLALEN if added to the shopper's planned list). The shopper then received either a \$1 or \$2 coupon for that category. Shoppers in the near condition received a \$1 or \$2 coupon for an unplanned category that was relatively close to their planned set (i.e., that would result in a small increase in ISLALEN if added to the shopper's planned list).

A total of 90 shoppers were intercepted at the store's main entrance over the course of two weekends in April 2012. We did not measure in-store travel distances in this study for two reasons. First, the far versus near coupon manipulation increases the path for redeeming shoppers by definition. Second, this store had not heretofore granted access for an academic study, so we were reluctant to risk being denied access by including path tracking in our proposal to store management.

Similar to Study 1, each shopper indicated her planned purchases from a list of product categories in the store and her shopping list was copied, if she had one. To provide an incentive yet mitigate a windfall effect, participants received a \$5 gift card after they finished shopping. To attribute any observed coupon effect to the coupon and not to shoppers randomly wandering around the store looking for their planned items, we included in the study only shoppers who indicated that they were relatively familiar with the store layout and that their mission was not a quick trip. Shoppers who qualified for the study completed an entry survey that assessed their planned items, which was used to determine the coupon (depending on the assigned experimental condition), as described previously. Shoppers were then given the coupon and released to do their shopping. After completing their shopping trip and checking out, participants returned to the experimenters to complete an exit survey in which they answered several demographic questions and completed the Rook and Fisher (1995) impulsivity scale. The experimenter copied each shopper's receipt and presented the \$5 gift card and cash for the coupon if he or she purchased the couponed category. Each shopper was then thanked and dismissed.

Results

Of the 90 shoppers who agreed to participate in the study, 1 had missing values and 4 other participants, all from the far condition, had an unusually high amount of unplanned spending (more than \$60); thus, we excluded them as outliers in the analysis.¹² Another four shoppers inadvertently received a coupon for a planned category and had to be dropped. This left 42 participants in the far condition and 39 participants in the near condition for analysis. Of the 42 participants in the far condition, 35 redeemed the coupon, and 37 of the 39 participants in the near condition redeemed the coupon.¹³ We included all participants in the analysis regardless of whether they redeemed the coupon, but the findings hold if we exclude the coupon nonredeemers. Our empirical analysis and previous simulations would lead us to believe that shoppers in the far condition would, on average, spend more on unplanned purchases than shoppers in the near condition. A coupon for an unplanned category that is farther away from the planned path encourages shoppers to travel more in the store, exposing them to more in-store stimuli and ideally spurring more unplanned spending.

The experimental results confirm our predictions. We conducted an analysis of variance with unplanned spending as the dependent variable and coupon path proximity, coupon amount, and their interaction as independent variables.¹⁴ As we expected, the main effect of coupon path proximity is significant ($F(1, 77) = 5.70, p = .02$). Neither the main effect of coupon amount nor the interaction between coupon path proximity and coupon amount is significant. The average amount of unplanned spending for shoppers in the far coupon condition (\$21.29) is much higher than for shoppers in the near coupon condition (\$13.83).

As a robustness check, we reran the analysis with the unplanned spending for the couponed category excluded. We obtained similar results; unplanned spending was \$20.14 in the far coupon condition versus \$12.50 in the near coupon condition ($p < .05$). These findings support our assertion that a mobile promotion strategy aimed at increasing in-store travel path length can be effective in increasing unplanned spending.

Discussion

We study whether wandering more in store has a direct, causal effect on the amount of unplanned spending, an aspect of in-store research that the academic literature has not fully addressed to date. As we explain in the beginning of the article, the main challenges are the difficulty of col-

¹²Including these outlier shoppers leads to even stronger results.

¹³Due to the study design, in which coupons were personally delivered to shoppers, the observed opt-in rate is much higher than would be expected if the coupons were delivered electronically. That said, the result that shoppers were much more likely to use the coupons for unplanned categories that entailed less additional travel provides a measure of face validity.

¹⁴An analysis of covariance with trip mission (stock up or fill in), number of planned items, and impulsivity revealed substantively identical results ($n = 68$ due to missing values).

lecting data and endogeneity issues due to potential omitted variables pertaining to in-store decision making, possible reversed causality/simultaneity, and measurement errors. To combat these issues, we collected in-store path data using person-level RFID tags and constructed a novel instrumental variable approach based on the lengths of reference paths that are determined by the store layout, a shopper's planned purchases, and assumptions about shopper's search strategy (TSA or 1SLA). Because reference paths are determined before the shopper begins his or her trip, our instruments temporally precede the dependent variable, allowing us to circumvent the endogeneity issues. Using 1SLALEN as an instrumental variable, we find that the elasticity of the amount of unplanned purchase on in-store path length is approximately 1.57, which is 57% higher than the estimate from OLS regression.

We then explore the effectiveness of two shopper marketing strategies—product category relocation and mobile promotion—on increasing unplanned spending. Our simulations suggest that relocating up to three product categories can increase unplanned spending by 7.2%, but targeted promotions of three product categories may increase unplanned spending by 16.1%. Because the two strategies are not mutually exclusive, retailers should consider using both strategies simultaneously to maximize revenue from unplanned purchases.

Through a follow-up field experiment, we further demonstrate that mobile promotions aimed at increasing travel distance significantly increase unplanned spending. Thus, another important takeaway from our research is that it demonstrates the potential efficacy of a mobile promotion strategy that is not being utilized at present. Most retailers have not yet launched a mobile shopping app or adopted turnkey solutions such as Modiv Media's scanners. Those that are currently using these technologies to send targeted promotions based on the shopper's location send promotions for *nearby* product categories. In sharp contrast, our research shows the benefit of sending promotions for categories that are *far* away from planned product categories.

We would be remiss not to note that although we have focused on smartphone-based mobile promotions here, there are other promising ways that can also deliver target promotions based on the shoppers' location.¹⁵ For example, traditional shelf-level instant coupons could be used to induce additional within-store travel by offering coupons for another part of the store. Alternatively, shopping basket data can be used to identify certain product combinations that are not typically purchased together. Using this prediction, an in-store promotion for one of these products can be offered at the location of another in that combination. In-store video screens have the ability to offer real-time messages targeted to a specific location within a store (Dukes and Liu 2011). By providing a message at one part of the store about products at another part of the store, shoppers may be induced to engage in additional within-store travel. Finally, loyalty card databases can be used to generate tar-

geted promotions by using shoppers' purchase history to predict which products are most likely to be unplanned on the current trip (Gilbride, Inman, and Stilley 2012) and offering a targeted promotion for an unplanned product in a seldom-visited part of the store.

While promising as a first step, our research has limitations that point to fruitful directions for further research. First, we have focused on increasing unplanned spending. It is important to consider where additional unplanned spending comes from: Does the additional unplanned spending represent truly incremental spending, or is it part of the unplanned spending "borrowed" from reduced planned purchases or even from future purchases? From the retailer's perspective, an increase in unplanned spending on the current trip tends to be preferred because it safeguards the retailer from losing the future purchase to a competitor. In our data set, roughly 70% of the planned purchases are actually purchased. Further research should be conducted to explore the relationships among unplanned purchases, planned baskets, and future purchases in more detail, perhaps using a panel data set based on a store loyalty card program.

Second, further research should also consider the "store choice" decision (Fotheringham 1988; Rhee and Bell 2002) in conjunction with the extent of unplanned spending. By lengthening the path that shoppers must walk to gather their planned purchases, some shoppers may view shopping in such a store as an unpleasant experience and refrain from shopping there in the future. This is particularly relevant for product relocation strategies; by forcing shoppers to roam all over the store, this strategy risks irritating shoppers. Similarly, if targeted mobile promotions are always offered for product categories that take consumers far from their planned shopping track, they may be perceived as irrelevant or even bothersome, resulting in a lower redemption rate. Thus, further research should delve deeper into the issue of how retailers can optimize the effectiveness of targeted promotion strategies while balancing store "shopability" perceptions.

Finally, further research should explore the extent to which our results generalize to grocery stores of different sizes or even to other types of stores and retail environments. For example, researchers may study the effect of increasing in-store travel distance on unplanned purchase in smaller stores such as convenience stores or in larger retail environments such as major shopping malls. These studies will further shed light on the drivers of unplanned purchase and allow retailers to further optimize their shopper marketing strategies.

Appendix

Solving the TSPLEN and 1SLALEN

We use the following example to illustrate how we solve for TSPLEN and 1SLALEN. Suppose a consumer plans to buy the following categories:

Category	Location (Store Zone)
Cooking oil	76
Eggs	20
Condiments	59
Fresh fish	15
Hardware	72

¹⁵We thank an anonymous reviewer for suggesting some of these approaches.

Thus, to obtain all his or her planned categories, the shopper must visit the entrance (zone 116); zones 76, 20, 59, 15, and 72; and the exit (zone 120). Table A1 shows the pairwise travel distances between these zones.

We define the TSP path (Hui, Fader, and Bradlow 2009b) as the path that begins at the entrance, connects all the zones where the shopper's planned items (zones 15, 20, 59, 72, and 76) are located, and ends at the checkout. To solve for TSPLN, we consider all the possible permutations of the order of visitation (e.g., 15 → 20 → 59 → 72 → 76, 20 → 59 → 15 → 76 → 72), and compute the total travel distance for each of the $k!$ permutations (where k is the number of planned zones). The order of visitation with the shortest distance is the TSP path, and its length is the TSPLN of that shopper. In the preceding example, the TSP path is Entrance → 59 → 15 → 20 → 72 → 76 → Exit, with a TSPLN of 570.04.

To compute 1SLALEN, we solve for the shopper's reference path using a 1SLA algorithm. We first locate the product zone that is closest to the entrance (zone 76); this is the first zone on the reference path. Next, from zone 76, we

look for the next product zone that is closest to zone 76, which is zone 72. Repeating the same procedure until the shopper reaches the exit, we find that the 1SLA reference path is Entrance → 76 → 72 → 20 → 59 → 15 → Exit, with a 1SLALEN of 639.56 (which is longer than the TSPLN, as expected, because the shopper is only looking forward one step at a time).

Note that in this case, the number of planned zones is relatively small ($k = 5$); this enables us to solve for the TSP path by investigating all possible permutations. When k is large ($k > 10$), we must solve for the TSP using a simulated annealing algorithm (e.g., Goffe, Ferrier, and Rogers 1994). Implementation details, including the C++ code, are available from the authors upon request.

We have verified that the simulated annealing converges to the globally optimal solution by running the simulated annealing with a slow exponential cooling schedule and repeating the algorithm ten times to ensure that they all converge to the same solution (details are available from the authors upon request). The 1SLA reference path can still be solved analytically regardless of the value of k .

TABLE A1
Pairwise Travel Distances Between Zones

	Entrance	Zone 15	Zone 20	Zone 59	Zone 72	Zone 76	Exit
Entrance	.00	169.95	238.86	129.07	164.44	126.51	76.01
Zone 15	169.95	.00	133.02	94.72	186.23	129.99	146.02
Zone 20	238.86	133.02	.00	109.79	87.21	126.77	169.88
Zone 59	129.07	94.72	109.79	.00	115.86	77.93	70.57
Zone 72	164.44	186.23	87.21	115.86	.00	75.30	88.64
Zone 76	126.51	129.99	126.77	77.93	75.30	.00	50.71
Exit	76.01	146.02	169.88	70.57	88.64	50.71	.00

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